# 數據、預測與 $R^2$

# Data, predictions and $R^2$

班別 / Class: _____ 組別 / Group: _____ 日期 / Date: _____

程式在 Colab：儲存副本，依次執行。將表格結果與觀察記在這裏。

Save a Colab copy and run the cells in order. Record results and observations here.

## 1 數據探索：外送距離與收費

Data exploration: delivery distance and fee

模擬紀錄；每點是一筆外送紀錄。執行第 1 節，看看散點圖。

Simulated records; each dot is one delivery. Run Section 1 and inspect the scatter plot.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 距離 / Distance (km) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 收費 / Fee (HK$) | 24 | 29 | 27 | 38 | 35 | 44 | 42 | 51 |

1. 距離增加時，收費大致怎樣變化？

As distance increases, how does the fee generally change?

2. 距離較遠，收費是否每次都較高？用兩筆紀錄說明。

Is the fee always higher for a longer distance? Use two records to explain.

## 2 線性回歸：溫習時間與成績

Linear regression: study time and exam score

執行第 2 節。將模型表的數值抄在下表；留意圖中的紫色預測點。

Run Section 2. Copy the model-table values below; find the purple prediction point.

| 項目 Metric | Colab 數值 Colab value | 意思 Meaning |
|---|---|---|
| n | _____ | 觀察數目 Number of observations |
| m | _____ | 每多一小時，估計分數的改變 Change in estimated marks per extra hour |
| c | _____ | 0 小時的估計分數 Estimated marks at 0 hours |
| $R^2$ | _____ | 直線解釋了多少數據差異 Share of variation explained by the line |
| 4.5 小時預測 Prediction at 4.5 h | _____ 分 / marks | 模型給出的估計，並非保證 An estimate, not a guarantee |

m、c 描述直線；不是準確度指標。所有成績爲模擬。$R^2$ 說明在背面。

m/c describe the line, not accuracy. Scores are simulated. See reverse for $R^2$.

Colab: tinyurl.com/26jdbxfb

### $R^2$（R square）：直線解釋了多少數據差異？

$R^2$: how much variation in these data does the line explain?

這裏的「差異」指各個 y 值有高有低，例如學生的成績不同。

Variation means that y-values differ, e.g. students have different scores.

本課的 $R^2$：在用來建立模型的數據上計算，值介乎 0 和 1。

Here, $R^2$ is measured on the data used to fit the line and ranges from 0 to 1.

接近 1：解釋較多差異；接近 0：直線解釋的差異較少。

Near 1: more variation explained. Near 0: little variation explained.

本組 $R^2 \approx 0.418$：解釋約 42% 的成績差異，不是 42% 的預測答對。

Here, $R^2 \approx 0.418$: about 42% of score variation, not 42% correct predictions.

新預測是否可靠？用未參與建模的新數據，比較預測與實際結果。

For new predictions, use data kept out of fitting; compare predictions with actual results.

## 3 比較 A、B、C：哪條線更能解釋自己的數據？

Compare A, B and C: which line explains its data better?

在第 3 節依次執行 A、B、C；各有一個程式格、21 點及相同座標範圍。

Run A, B and C in Section 3. Each has one code cell, 21 dots and the same axes.

先看圖，再記下 $R^2$。第 1 名代表解釋自己數據差異的比例最高。

Inspect each graph and record $R^2$. Rank 1 means the most variation explained.

| 例子<br>Example | $R^2$（抄至 3 位小數）<br>$R^2$ (copy 3 decimal places) | 排名：1 / 2 / 3<br>Rank: 1 / 2 / 3 |
|---|---|---|
| A | _____ | _____ |
| B | _____ | _____ |
| C | _____ | _____ |

1. 第一名的圖中，你看到甚麼支持這個排名？

What do you see in the top-ranked graph that supports your ranking?

2. 高 $R^2$ 是否保證能準確預測新數據？還需要做甚麼檢查？

Does high $R^2$ guarantee accurate new predictions? What check is still needed?

三組都是模擬數據。隨機數據也可能碰巧有小幅趨勢。 / Random data can show a chance trend.